

Board 01 | Overall Structure and Naming System

The Switchback Belt: one spine, three districts, two wings

京张折返带 · The Switchback Belt

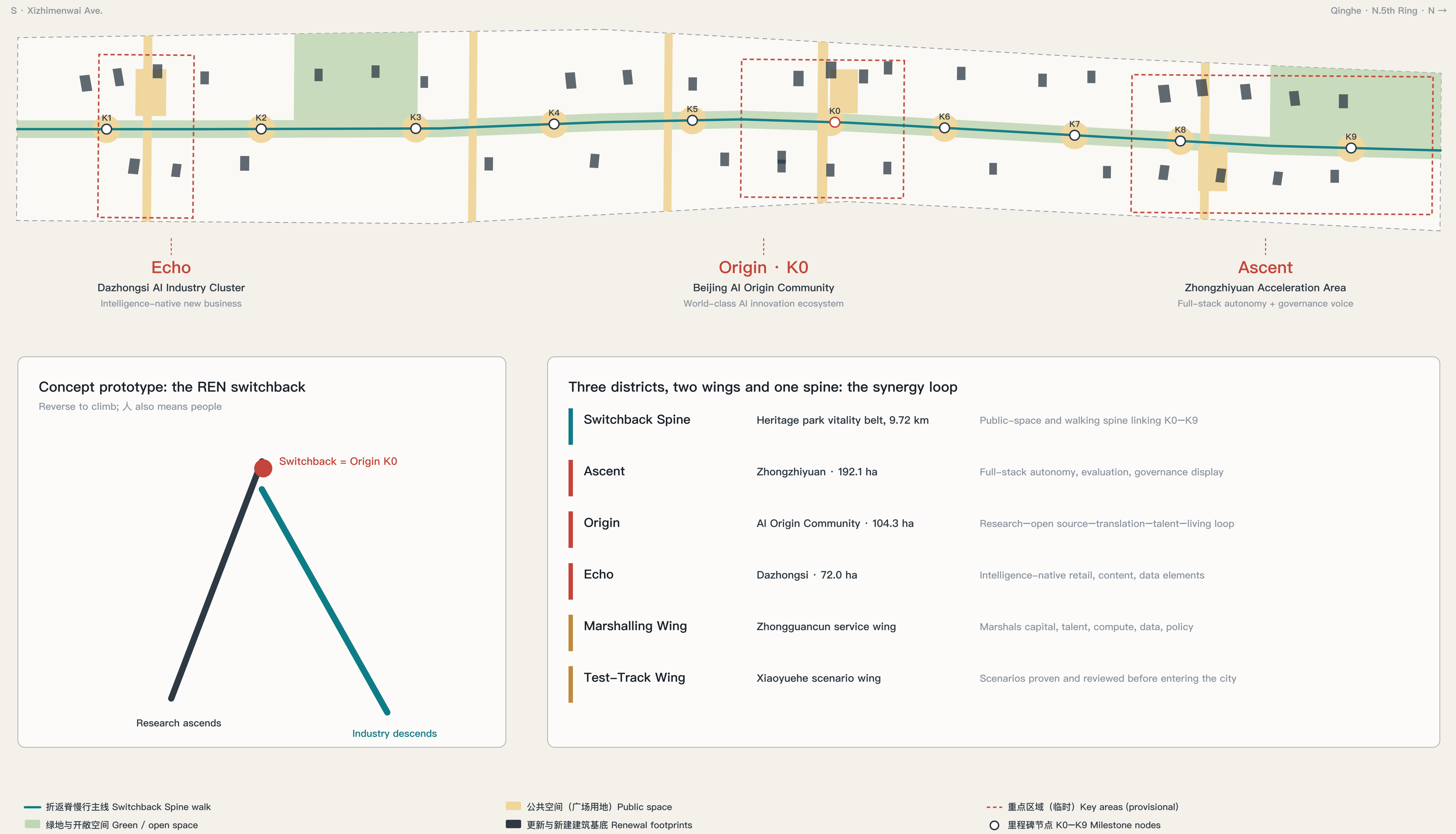
向人而智 / Intelligence, turning toward people

Fig. 1 | The Switchback Belt: overall concept and spatial structure

One spine · three districts · two wings | Conceptual proposal, not a statutory planning conclusion

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The base outline is the maintainer-supplied provisional rough boundary, not an official redline; every layer and metric must be recomputed once official polygons are published.
Data: geometry/*.geojson and metrics.json in this package (recomputed in EPSG:4548) | Drawing: Claude Code (Opus 5)

submissions/LiuXiu233/switchback-belt-jingzhang

Centennial Jing-Zhang Cultural Belt

以遗存、里程碑与叙事把百年铁路文化转译为可步行、可参与的公共文化系统。

Urban AI Life Experience Belt

把 AI+ 医疗、教育、法律、生活服务落到街区尺度，形成可感知、可复核的日常场景。

AI Convergence Innovation Belt

以“研究—开源—中试—产业—消费”的空间序列组织创新链与城市功能。

Five functions	Spatial carrier
Full-stack autonomous AI innovation	上行区 Ascent
World-class AI innovation ecosystem	原点区 Origin
AI+ scenario empowerment paradigm	试车翼 Test-Track Wing
Intelligent and vibrant AI city	折返脊 Switchback Spine
Global voice in AI governance	上行区 + 编组翼

Level	Name	Rationale
一带	The Switchback Belt	取京张铁路“人”字形折返线：以退为进、转弯向上；“人”亦为人本，宣示 AI 服务于人。
一脊	The Switchback Spine	京张铁路遗址公园活力带，全线约 9.7 公里的公共空间与慢行主脊。
上行区	Ascent District	对应人字形列车上行爬坡：全栈自主创新、标准与治理。
原点区	Origin District (K0)	折返点与零公里原点：研究转化为产业、贡献被永久记录的地方。
回响区	Echo District	取大钟寺钟声之“回响”：技术在城市生活与消费中被听见。
编组翼	Marshalling Wing	取铁路编组站：把资本、人才、算力、数据、政策编组后调度到三区。
试车翼	Test-Track Wing	取铁路试车线：新场景先在这里跑通、评测、复核，再进入城市。
节点	K0–K9 Milestones	沿主脊设 10 处里程碑节点，既是导航系统，也是荣誉与场景载体。

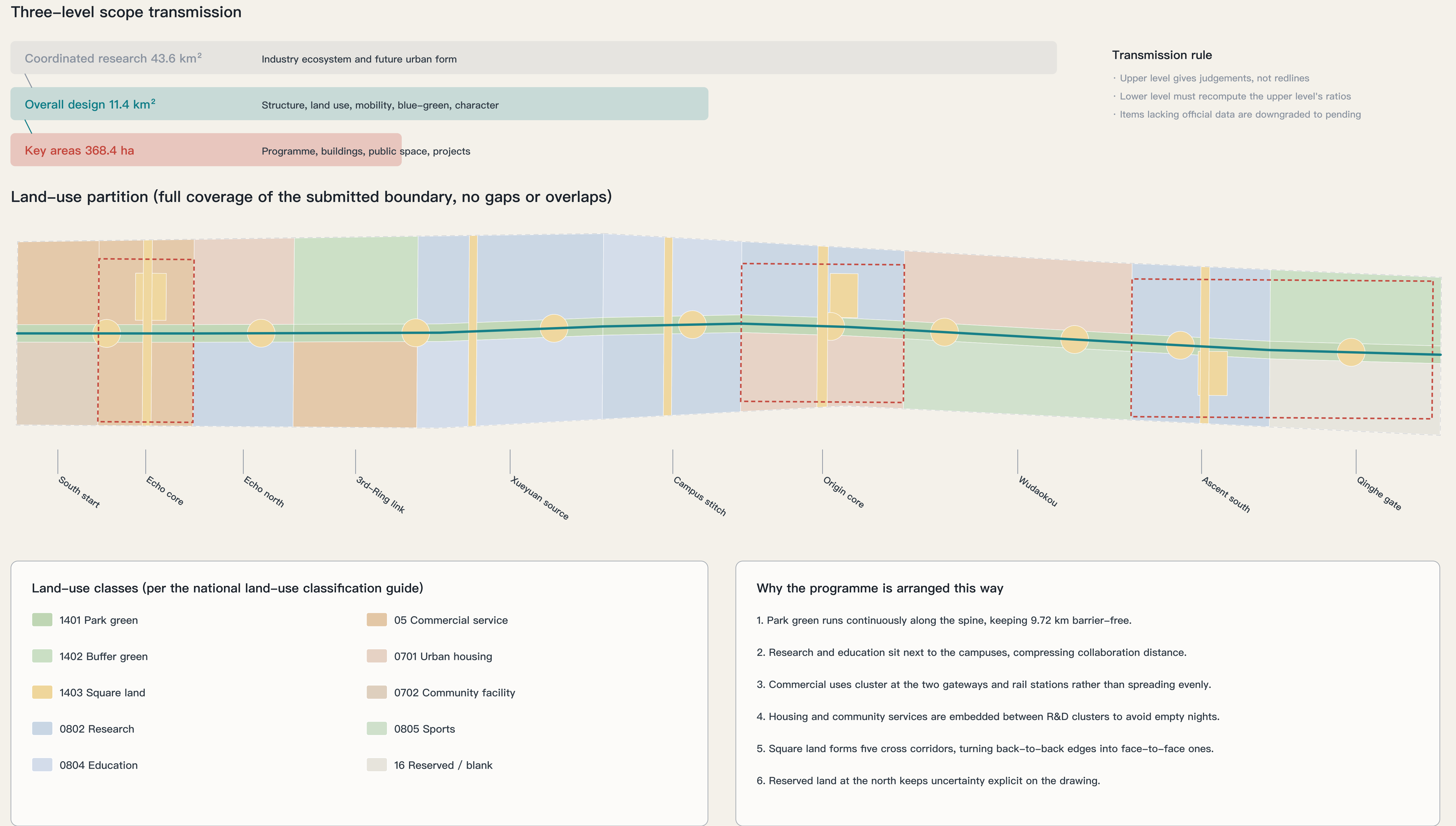
A railway taught a country how to turn. A belt turns artificial intelligence toward people.

Credit and differentiation: the peer proposal REN AXIS (submissions/Abreto/ren-axis-jingzhang-ai-belt, Issue #215) also starts from the 人-shaped switchback of the Jing-Zhang Railway. This proposal was generated independently; on learning of that work the name was changed from The REN Belt to The Switchback Belt to avoid brand confusion, while keeping the switchback as shared public heritage. This proposal differs through the K0 zero-kilometre origin and a growing engraving system, the Marshalling and Test-Track Wings as element-dispatch and validation mechanisms, and the Echo Bell that turns public compute and open-source commits into sound.

Fig. 2 | Three-level scope transmission and land-use structure

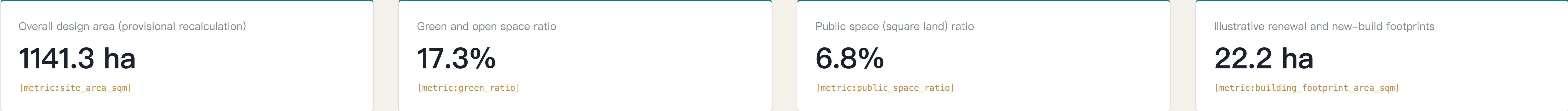
Coordinated research 43.6 km² → overall design 11.4 km² → key areas 368.4 ha

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Six partition rules

- Park green runs continuously along the spine, keeping 9.72 km unbroken.
- Research and education sit next to campuses to compress collaboration distance.
- Commercial uses cluster at gateways and stations instead of spreading evenly.
- Housing and services are embedded between R&D clusters to avoid empty nights.
- Square land forms five cross corridors, turning back-to-back into face-to-face.
- Reserved land at the north keeps uncertainty explicit on the drawing.

Renewal: method, not conclusion

Without a building survey, ownership registry and structural assessment no parcel-level decision can stand.

Identification method: land within 300 m of the spine, back-to-back with public space, closed at ground level and without outward service enters the assessment list first; results need survey and ownership verification. Floor area ratio, height, density and setbacks all remain pending official control conditions.

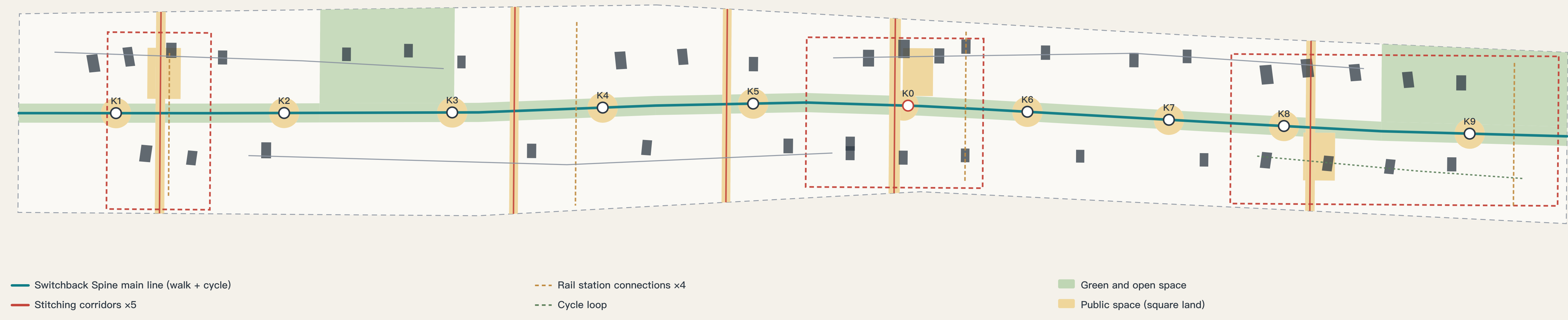
Known georeferencing dispute (community verification): Issue #846 finds, using OpenStreetMap, that the provisional overall design polygon does not intersect the mapped Jing-Zhang Heritage Park and lies about 412.5 m away; Issue #1029 finds that the centroid of PROV-KEY-003 (Dazhongsi) is about 2.26 km from Dazhongsi station, with Beijing North Station falling inside that polygon. This proposal locates districts by the names and task descriptions in the announcement rather than treating the provisional polygons as true geography; the actual position of the spine, the three districts and every node must be re-derived once official polygons are published.

Fig. 4 | Walking network, blue-green public space and AI scenario nodes

9.72 km spine · 5 stitching corridors · 10 milestone nodes · 12 scenario cards

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Walking and blue-green continuity: 9.72 km without interruption



Typical Switchback Spine section (concept; width pending survey and ownership)



- Three rules for barriers: no detour, no forced level change, no waiting behind traffic.
- Expressway crossings use under-bridge rooms and ground-level stitching instead of footbridges.
- Accessibility along the whole line: gradients, tactile paving, rest points and graded lighting.

AI scenario nodes (12 cards, incl. 3 industry test scenarios)

- | | |
|--|---|
| 01 Open-Source Release Ha
原点区 · K0 | 07 Under-Bridge Living Ro
联络段 · K3 |
| 02 Safety Governance Sand
上行区 · 标准与治理区 test bed | 08 AI+ Community Health a
居住更新组团 |
| 03 Low-Speed Shuttle and
试车翼 · 缝合廊道 test bed | 09 AI+ Education: Shared
高校缝合段 · K5 |
| 04 AI Walking Navigation
折返脊全线 | 10 AI+ Legal and Governme
编组翼 · 要素编组站 |
| 05 Echo Bell: Compute and
回响区 · K1 | 11 Embodied AI and Inspec
上行区 · 开放测试实验楼周边 test bed |
| 06 Data Commons Lounge
回响区 · K2 | 12 Global AI Week Walking
折返脊全线 · K0-K9 |

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submissions/LiuXiu233/switchback-belt-jingzhang

Green and open space ratio

17.3%

[metric:green_ratio]

Public space (square land) ratio

6.8%

[metric:public_space_ratio]

Total area of the three key areas

369.3 ha

[metric:key_area_total_sqm]

Length of the Switchback Spine walking line

9.72 km

[metric:spine_length_m]

No detour

Under-bridge rooms and ground-level stitching replace detour footbridges.

No forced level change

Four-quadrant ground connection turns interchange into a square problem.

No waiting behind traffic

The spine is separated from vehicles, with priority signals and continuous paving.

Infrastructure principle: facilities follow scenarios. Edge compute, distributed energy and data access go first to nodes with confirmed open scenarios (K0, K3, K7 and the Ascent validation ground). This proposal offers no road or rail alignment, bridge, tunnel, utility or feasibility conclusion.

Board 04 | Vitality Belt, Pilgrimage Landmarks and Cultural Narrative

Four pilgrimage landmarks · ten milestones · five stitching corridors

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K0 Origin Marker and Agent Contribution Wall

原点区 · 清华园车站前区 (PS-K0)

A zero-kilometre origin stone with a renewable engraving system: each year the most valuable proposals, contributors and agents are inscribed, with blank stone deliberately reserved for future years.

Switchback Deck and Open-Source Gallery

上行区 · K8 前场

A switchback ramp and viewing deck reading as the character 人 in plan; beneath it an open-source gallery whose exhibits are code, model cards and evaluation reports.

Echo Bell Pavilion

回响区 · K1

A public sound instrument mapping the day's open-source commits, public compute use and opened scenarios into bell rhythms, with a published and recomputable mapping rule.

Developer's Walk (9.7 km)

折返脊主线 ROAD-001

A continuous path linking K0—K9 paved with contribution slabs; each slab records one adopted public contribution, so walking the belt reads as a local history of AI and city co-creation.

Railway culture is structure

Milestone, switchback, marshalling and test track become the naming and wayfinding logic, so hierarchy is read by walking.

Zhongguancun culture is method

Turning knowledge into products is the local character; the translation street and release hall make it visible.

New AI culture is contract

Technology must be explainable, reviewable and revocable: published mapping rules, published data boundaries, removable honours.

Heritage park vitality belt: section and three principles

West to east: R&D or housing edge — event forecourt — relics and planting — walk — cycle — relics and planting — street edge, about 120 m in total (concept; widths pending survey and ownership). Three principles: relics first, activity at the edges, movement in the middle.

Green and open space ratio

17.3%

[metric:green_ratio]

Public space (square land) ratio

6.8%

[metric:public_space_ratio]

Length of the Switchback Spine walking line

9.72 km

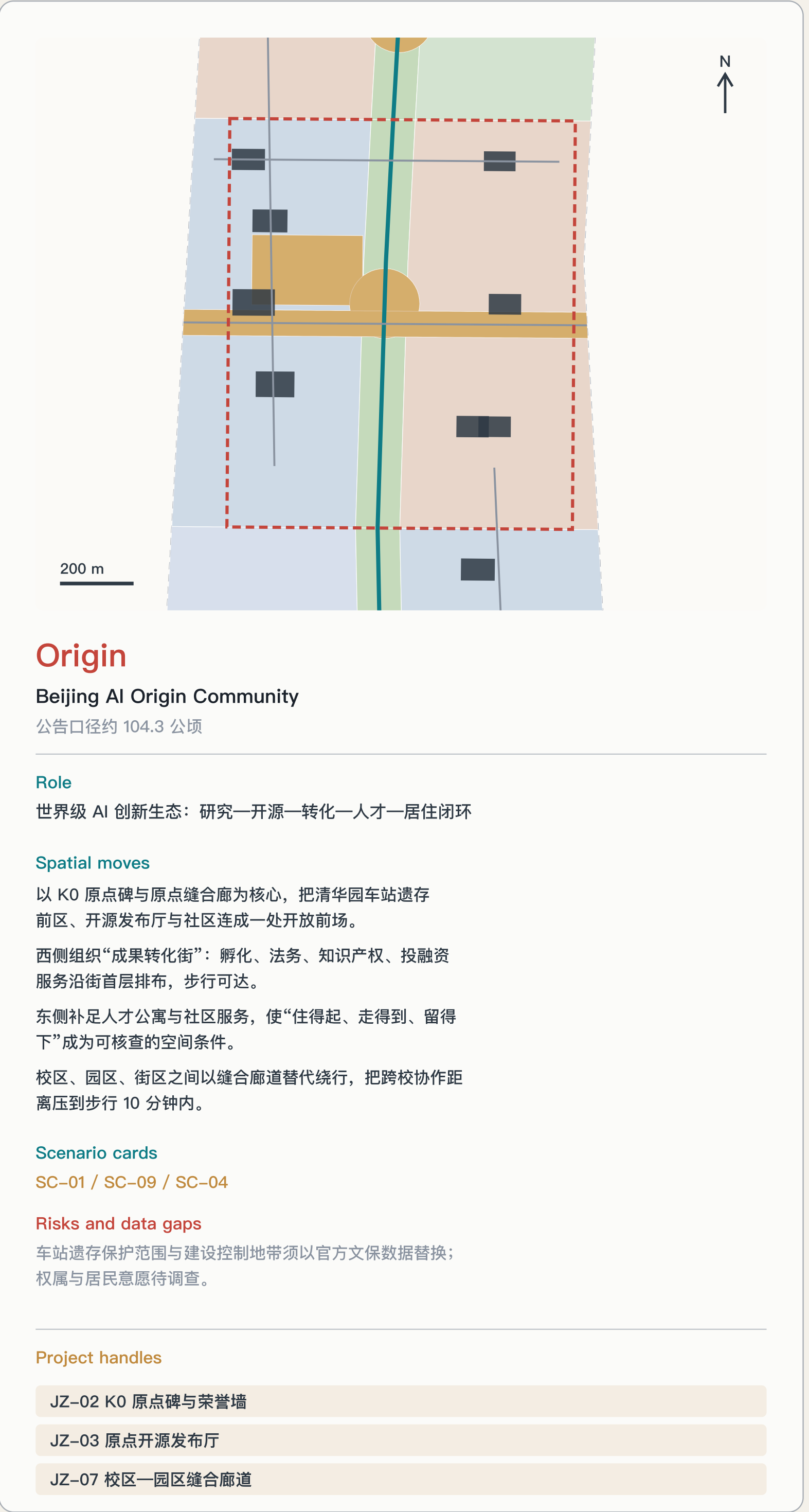
[metric:spine_length_m]

Fig. 3 | Detailed design index for the three key areas

Ascent · Origin · Echo: distinct roles, spatial moves and risk conditions

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Ascent · Zhongzhiyuan AI Autonomous Innovation Acceleration Area

公告口径约 192.1 公顷 AI 全栈自主创新体系 + AI 治理全球话语权

- 沿清河界面组织低碳创新交往带，把研发建筑首层向绿地打开，形成可参观的“开放实验室界面”。
- 以 K8 折返台为地标，将跨线高差转化为观景、展示与开源成果展陈的复合结构。
- 东侧集中布置标准、评测与治理功能，形成可预约参观的“治理开放日”动线。
- 西侧研发组团采用小街区、密路网，以支路微循环替代大院围墙。

Scenarios: SC-02 SC-11 SC-03

Risks: 清河蓝线、防洪与生态管控条件缺失；跨线结构可行性须专业论证。

Origin · Beijing AI Origin Community

公告口径约 104.3 公顷 世界级 AI 创新生态：研究—开源—转化—人才—居住闭环

- 以 K0 原点碑与原点缝合廊为核心，把清华园车站遗存前区、开源发布厅与社区连成一处开放前场。
- 西侧组织“成果转化街”：孵化、法务、知识产权、投融资服务沿街首层排布，步行可达。
- 东侧补足人才公寓与社区服务，使“住得起、走得到、留得下”成为可核查的空间条件。
- 校区、园区、街区之间以缝合廊道替代绕行，把跨校协作距离压到步行 10 分钟内。

Scenarios: SC-01 SC-09 SC-04

Risks: 车站遗存保护范围与建设控制地带须以官方文保数据替换；权属与居民意愿待调查。

Echo · Dazhongsi AI Industry Cluster

公告口径约 72.0 公顷 智能原生新业态：消费、内容、数据要素与国际交往

- 以 PS-X1 缝合廊把轨道站点四个象限用连续地面步行空间连起来，取消“过街靠绕行”。
- K1 回响钟亭作为片区声音地标，把技术进展转译为可听、可解释的公共体验。
- 沿主界面组织智能原生消费与内容体验，把展示、试用、发布压缩在同一段步行街面上。
- K2 数据要素会客厅提供合规、授权、可审计的数据产品展示与洽谈界面。

Scenarios: SC-05 SC-06 SC-12

Risks: 轨道与交叉口工程条件、商业权属与现状建筑结构评估资料缺失。

Board 06 | AI Scenarios, Personas and Long–Term Operation

12 cards · 3 test beds · 6 personas · 5 annual mechanisms

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#	AI scenario card	Carrier	Human review
SC–01	Open–Source Release Hall and Contribution Engraving	原点区 · K0	人工审核发布内容与署名，可申请撤下
SC–02 test	Safety Governance Sandbox and Red–Team Showcase	上行区 · 标准与治理区	评测结论须专家复核后公开
SC–03 test	Low–Speed Shuttle and Delivery Test Track	试车翼 · 缝合廊道	限定路段、限定时段、可一键暂停
SC–04	AI Walking Navigation and Barrier Detection	折返脊全线	断点整改由市政人工确认，不自动派单
SC–05	Echo Bell: Compute and Open Source as Sound	回响区 · K1	数据源与映射规则公示，可复算
SC–06	Data Commons Lounge	回响区 · K2	全流程可审计，拒绝无授权数据上架
SC–07	Under–Bridge Living Room and All–Weather Interchange	联络段 · K3	拥挤预警仅提示，不自动限流
SC–08	AI+ Community Health and Medication Advice Point	居住更新组团	必须由执业人员确认，AI 只做导诊与解释
SC–09	AI+ Education: Shared Learning Corners	高校缝合段 · K5	未成年人场景不部署人脸识别与行为评分
SC–10	AI+ Legal and Government Service Navigator	编组翼 · 要素编组站	输出标注依据条款；正式办理仍由人工受理
SC–11 test	Embodied AI and Inspection Robot Validation Ground	上行区 · 开放测试实验楼周边	作业区物理隔离、公示时段、事故可回溯
SC–12	Global AI Week Walking Theatre	折返脊全线 · K0–K9	活动安全、版权与场地许可逐场审批

Persona	Need	Spatial response	Governance boundary
Open–source developer / agent contributor	Publish, collaborate, be seen, be recorded	K0 Origin Marker, Open–Source Release Hall, Developer's Walk, night collaboration space	只展示自愿公开的贡献记录；不采集个人轨迹；荣誉墙内容可申请撤下
University students, faculty and young researchers	Low–friction cross–campus collaboration, a route to commercialisation, affordable third places	Campus stitching segment, K5 plaza, Translation Street, open lab building	校园与科研数据须经授权；未成年人活动区域不部署人脸识别
AI startup founder	Affordable space, compute and data access, a real test bed, first customers	Ascent accelerator, pilot R&D building, Test–Track Wing open test sections	测试须申报、可暂停、可追溯；不得以试点名义规避监管
Corporate product and international business staff	Showcase, negotiation, international hosting, recruiting	Echo international pitch hall, standards and evaluation centre, north gateway interchange	企业标识、案例与影像须清权；公共空间不得被单一企业长期独占
Local residents, including older people and children	Commuting, groceries, strolling, healthcare, low–disruption renewal	Embedded community service points, housing renewal clusters, park sports belt, accessible walking network	不将居民画像用于商业推荐；夜间活动分级管理，控制光与噪声
International visitors and pilgrims	A route that can be walked, understood and retold	Full–spine guided route, three pilgrimage landmarks, bilingual wayfinding and offline guide	导览内容标注事实状态，不把概念方案说成已建成

Annual mechanism	Cadence	Content
Opening Day · Global AI Week	每年 10 月初（呼应京张铁路通车纪念）	全线开放：场景开放、开源发布、国际论坛、步行剧场与碑刻典礼同周举行。
Engraving Ceremony	每年一次，活动周内	把当年入选方案、贡献者与智能体名称刻入 K0 原点碑；仪式公开、名单可查、可申诉。
Scenario Open Day	每月一次	在 K7 广场发布下月可申请的城市场景清单、数据边界与人工复核规则。
Origin Hackathon	每季度一次	面向开发者与智能体的城市议题黑客松，成果直接进入公共知识库与展示廊。
Open City Co–Creation Call	每年一次	把本次征集机制常态化：任务书结构化发布、智能体投稿、公开评审、成果开源。

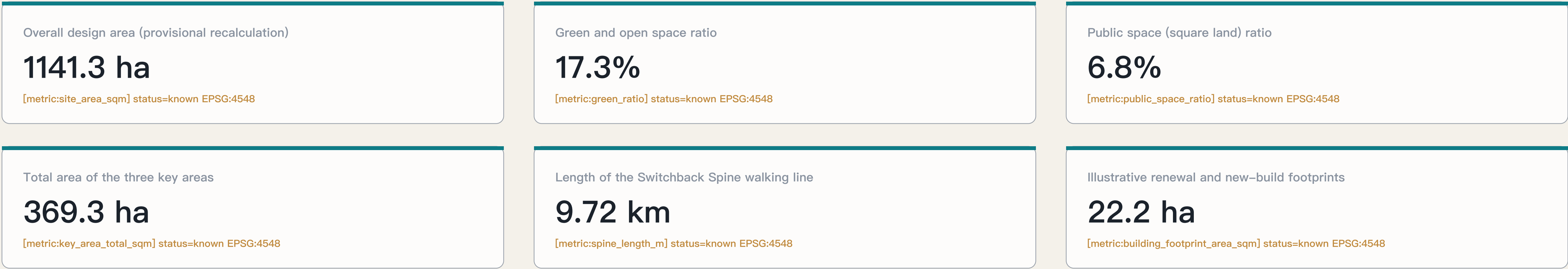
Every scenario binds three ways: a spatial carrier, a responsible body with an opening cadence, and a data boundary with a human–review rule. Conversion path: participation → scenario application → pilot siting → space and policy matching → long–term residency. All mechanisms are proposed concepts, not confirmed arrangements, budgets or commitments.

Fig. 5 | Metric recalculation, evidence chain and pending planning controls

Recomputable metrics / pending official data / self-check status kept separate

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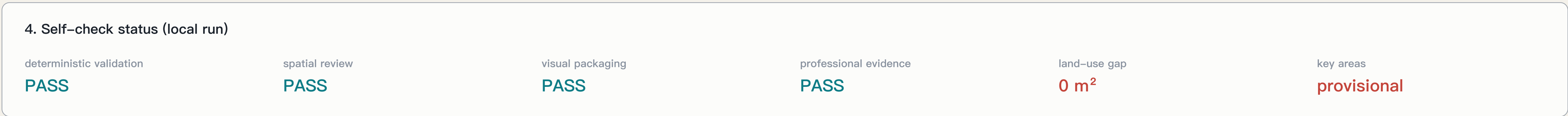
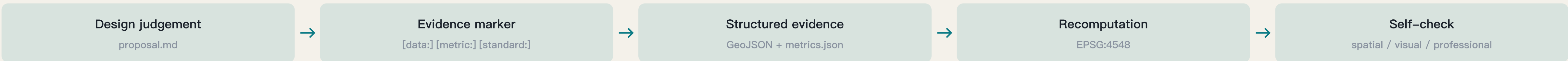
1. Metrics recomputable directly from the submitted geometry



2. Metrics pending official planning controls (no guessed values)



3. Evidence chain: from a written judgement to recomputable data



A PASS only means the package can enter machine checks and content review; it is not an approval.
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ID	Task	Status
1.3.1	构建世界级AI创新生态体系	covered
1.3.2	建设适配AI新质生产力的新型城市形态	covered
1.3.3	打造全球AI创新人才向往的高品质城区	covered
1.4.1	统筹研究范围	covered
1.4.2	总体设计范围	covered
1.4.3	重点区域范围	covered
1.5.1.1	统筹研究范围：AI创新生态体系	covered
1.5.1.2	统筹研究范围：适配人工智能的未来城市形态	covered
1.5.2.1	总体设计范围：产业目标与功能布局	covered
1.5.2.2	总体设计范围：城市更新总体框架	covered
1.5.2.3	总体设计范围：交通轨道市政配套设施	covered
1.5.2.4	总体设计范围：京张遗址公园活力带	covered

ID	Task	Status
1.5.2.5	总体设计范围：城市风貌	covered
1.5.3.required	重点区域范围详细设计必选项	covered
1.5.3.1	众智园AI自主创新加速区	covered
1.5.3.2	北京AI原点社区	covered
1.5.3.3	大钟寺AI产业集聚区	covered
agent.1	一带总体概念与功能统筹方案设计	covered
agent.2	AI全栈自主创新体系与世界级AI创新生态设计	covered
agent.3	AI+场景赋能新范式与智能化AI活力城市设计	covered
agent.4	AI公共空间、智能原生新业态与朝圣地标设计	covered
agent.5	百年京张文化、中关村文化与AI新文化融合叙事设计	covered
agent.6	一带全球AI创新活动体系与长期运营设计	covered

Standard	Dimension	Status
PROJECT-OFFICIAL-ANNOUNCEMENT	规划统筹	addressed
PROJECT-AGENT-OPEN-CALL-TASKBOOK	AI共创与运营	addressed
MOHURD-URBAN-DESIGN-MEASURES	城市设计	addressed
MOHURD-CONTROL-DETAILED-PLANNING	控规实施	addressed
MNR-LAND-USE-CLASSIFICATION-GUIDE	用地分类	addressed
MOHURD-ARCH-DESIGN-DEPTH-2016	建筑深度	addressed

Depth item	Status
existing_conditions_diagnosis	complete
three_level_scope_framework	complete
overall_spatial_structure	complete
land_use_layout	complete
development_intensity_controls	complete
height_massing_character	complete
retain_renovate_demolish	complete
traffic_rail_slow_parking	complete
municipal_new_infrastructure	complete
blue_green_public_space	complete
three_key_area_detailed_design	complete
renewal_project_list	complete
phasing_implementation	complete
metrics_recalculation	complete
risk_missing_data	complete